Syllabus Year: 2024-2025

CHRIST (Deemed to be University), Bengaluru - 560029

END-SEM EXAMINATIONS - APRIL 2025

Bachelor of Computer Applications SEMESTER - IV

Course Code: BCA201-4 **Max Marks: 100**

Course Name: GRAPH THEORY **Duration: 03:00**

General Instructions:

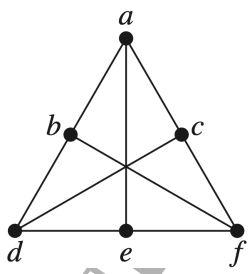
- All rough work should be done in the answer script. Do not write or scribble in the question paper except your register number.
- Verify the Course code / course title and number of pages of questions in the question paper.
- Make sure your mobile phone is switched off and placed at the designated place.
- Malpractices will be viewed very seriously.
- Answers should be written on both sides of the paper in the answer book. No sheets should be detached from the answer book.
- Answers without the question numbers clearly indicated will not be valued. No page should be left blank in the middle of the answer book.

Course Outcome: Students will be able to:
CO1: Enhance research, inquiry and analytical thinking abilities. CO2: Improve proof writing skills.
CO3: Apply the basics of Graph Theory in solving practical problems

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Section A **3*8=24**

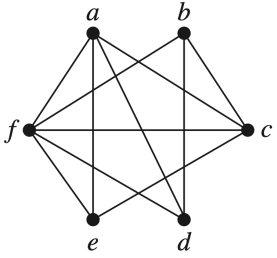
1)	Can a simple graph exist with 15 vertices each of degree 5? Give reasons for the same.	**3**	(CO3, L4)
2) a)	Draw these graphs. (a) Q_4 (b) $K_{1,8}$ (c) W_8	**3**	(CO3, L3)
	OR		
b)	If G is a simple graph with 20 edges and complement of G with 16 edges, how many vertices does G have?	**3**	(CO3, L3)
3)	How many vertices does a regular graph of degree four with 10 edges have?	**3**	(CO3, L4)
4)	Suppose that a connected planar graph has eight vertices, each of degree three. Into how many regions is the plane divided by a planar representation of this graph?	**3**	(CO1, L3)
5) a)	Determine whether the following graph is planar. If so draw it so that no edges cross.	**3**	(CO1, L4)



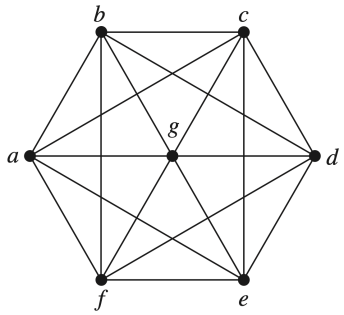
OR

b)	Define chromatic number of a graph. What is the chromatic number of the following graph?	**3**	(CO1, L4)

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6) State Kuratowski's theorem. Use Kuratowski's theorem determine whether the following graph is planar. **3** (CO1, L3)



7) Define a spanning tree. Find a spanning tree of $K_{4,4}$. **3** (CO2, L3)

8) a) How many edges will have to be removed from complete graph K_n to get its spanning tree? Justify your answer. **3** (CO2, L2)

OR

b) How many edges must be removed from a connected graph with n vertices and m edges to produce a spanning tree? **3** (CO2, L2)

Section B **8*7=56**

9) (i) Define adjacency matrix and incidence matrix. Represent W_5 and $K_{2,3}$ with adjacent and incident matrices. **8** (CO3, L3)

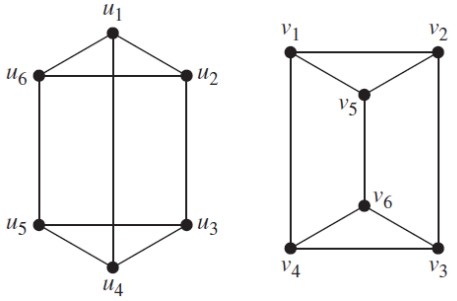
(ii) List out any four properties each of the adjacency and incidence matrices of a graph.

10) a) Graph G is defined as follows: The set of vertices is $V = \{0, 1, 2, 3, 4, 5, 6, 7, 8\}$, and two vertices u and v are adjacent if $|u - v| \in \{1, 4, 5, 8\}$. Draw and determine the order and the size of the following subgraphs of G: **8** (CO3, L3)

i) The subgraph induced by even vertices.
ii) The subgraph induced by odd vertices.
iii) The subgraph induced by the set $\{0, 1, 2, 3, 4\}$.
iv) A spanning subgraph with as many edges as possible but without cycles.

OR

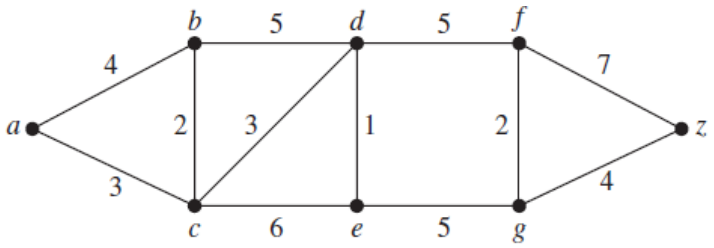
b) Explain the isomorphism of graphs. Examine whether the following graphs are isomorphic. Justify. **8** (CO3, L3)



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11) Determine whether the complement of Q_3 is planar. If so give a planar representation. **8** (CO1, L3)

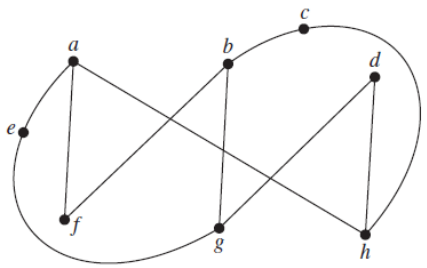
12) Determine the shortest path from the vertex a to the vertex z using Dijkstra's algorithm. **8** (CO1, L3)



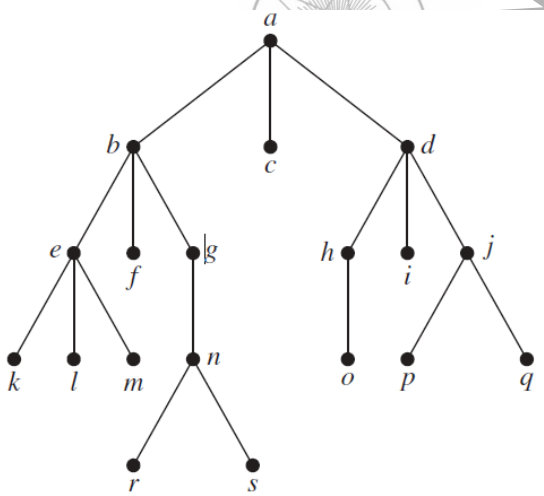
13) a) Determine the chromatic number of the following graph i) $K_{m,n}$ ii) $Q_n; 1 \leq n \leq 3$ iii) W_n. **8** (CO1, L2)

OR

b) What do you mean by subdivision of an edge? Define what is homeomorphism between a pair of graphs. Determine whether the following graph is homeomorphic to $K_{3,3}$. **8** (CO1, L2)



14) In which order are the vertices of the ordered rooted tree visited using a postorder traversal? **8** (CO2, L4)



15) a) Construct the ordered rooted tree whose preorder traversal is $a, b, f, c, g, h, i, d, e, j, k, l$, where a has four children, c has three children, j has two children, b and e have one child each, and all other vertices are leaves. **8** (CO2, L3)

OR

b) Represent the expressions (x + xy) + (x/y) and x + ((xy + x)/y) using binary trees. Write these expressions in **8** (CO2, L3)

a) prefix notation.

b) postfix notation.

c) infix notation.

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Section C

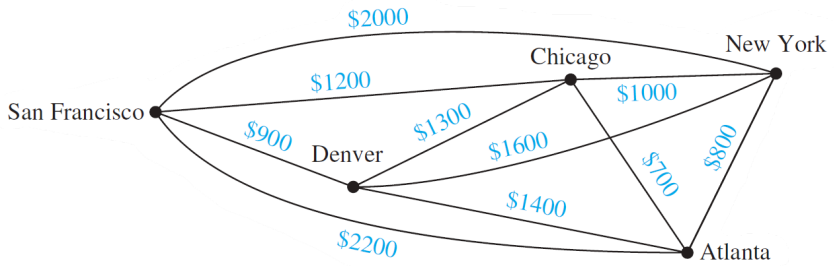
10*2=20

16) a) Draw all the non-isomorphic regular subgraphs on 6 vertices of K_6 and determine their chromatic number. **10** (CO2, L2)

OR

b) Draw all the non-isomorphic planar subgraphs on 4 vertices of K_4. **10** (CO2, L2)

17) Use Kruskal's algorithm to design a minimum-cost communications network connecting all the computers represented by the following graph. **10** (CO2, L4)



Revised Bloom's Taxonomy (RBT) Levels		
L1 - Remember	L2 - Understand	L3 - Apply
L4 - Analyse	L5 - Evaluate	L6 - Create

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